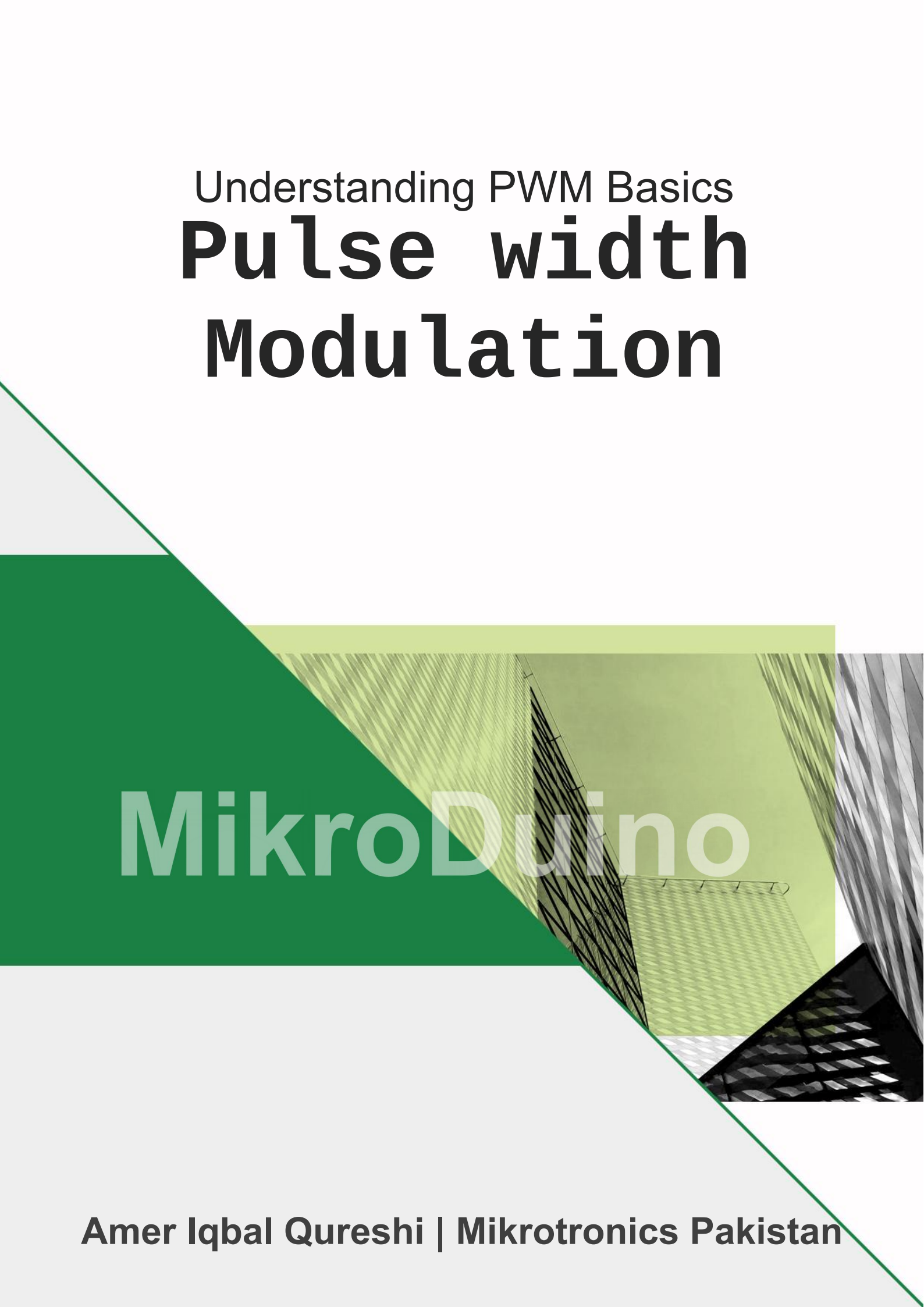


Understanding PWM Basics

Pulse width Modulation



MikroDuino

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Pulse Width Modulation (PWM)

Pulse Width Modulation (PWM) is one of the most important and widely used techniques in modern electronics and embedded systems. It allows a digital device, such as a microcontroller, to precisely control the amount of electrical power delivered to a load by rapidly switching its output between ON and OFF states. Instead of varying the output voltage directly, PWM changes the proportion of time that the signal remains HIGH during each switching cycle, thereby controlling the average power supplied to the connected device. This simple yet powerful technique makes it possible to control the brightness of LEDs, regulate the speed of DC motors and cooling fans, position servo motors, generate audio tones, create analog-like voltages, drive switching power supplies, and even synthesize complex waveforms. Because PWM combines high efficiency, precise digital control, and minimal power loss, it has become a fundamental feature of virtually every modern microcontroller and an indispensable tool for embedded systems engineers. Understanding PWM is therefore essential for anyone interested in designing intelligent, energy-efficient, and high-performance electronic systems.

Part I – Understanding Pulse Width Modulation (PWM)

Making the Impossible Possible

Imagine that you are standing in front of a room with a single light switch. The switch has only two positions:

- ON
- OFF

There is no middle position. The light is either fully bright or completely dark. Now suppose someone asks you to make the lamp glow at **half brightness**. At first glance, this seems impossible. After all, the switch cannot produce "half ON."

But what if you could flip the switch **hundreds or even thousands of times every second**?

If the lamp is turned ON for half of the time and OFF for the other half, the human eye no longer sees individual flashes. Instead, it perceives the light as glowing continuously—but only at about **50% brightness**.

Nothing magical has happened. The lamp is still receiving only full voltage or zero voltage. The only difference is **how long the voltage remains ON during each cycle**.

This simple but extremely powerful idea is called **Pulse Width Modulation**, commonly abbreviated as **PWM**. Rather than changing the voltage itself, PWM changes the **width of each pulse**.

This technique allows digital electronics to imitate analog behavior with remarkable precision.

From Digital to Analog

One of the first things students learn about digital electronics is that a digital output pin has only two possible states.

LOW = 0 Volts

HIGH = 5 Volts (or 3.3 Volts)

Unlike laboratory power supplies, a microcontroller cannot normally produce 1V, 2.5V, or 4V directly on its digital output pins. The output is always either:

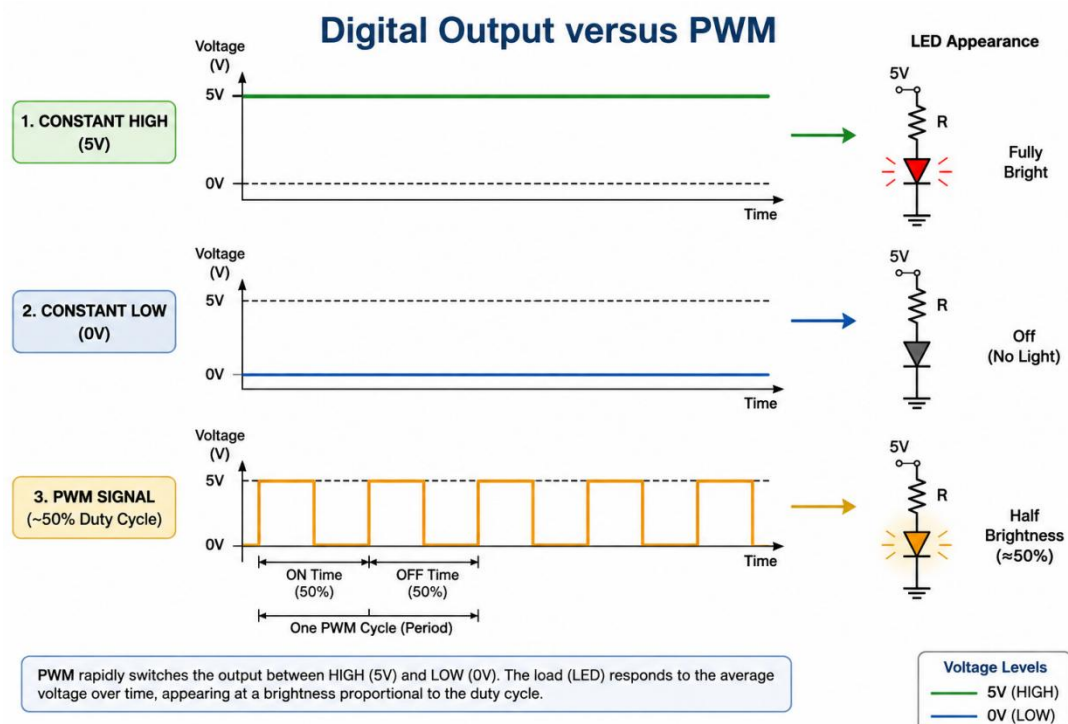
HIGH or LOW

This seems like a severe limitation.

How can a microcontroller:

- Dim an LED?
- Control the speed of a DC motor?
- Generate audio?
- Produce analog voltages?
- Regulate power delivered to a heater?

The answer lies in **Pulse Width Modulation**. Instead of trying to generate intermediate voltages, the microcontroller rapidly switches the output between HIGH and LOW. If this switching is fast enough, the load responds to the **average energy** rather than to the individual pulses.



Understanding the Average Effect

Consider the following PWM waveform.



During every cycle:

- The signal remains HIGH for half the time.
- The signal remains LOW for the remaining half.

Although the voltage is still switching between 0V and 5V, the average voltage becomes approximately:

Average Voltage

= Duty Cycle $\times$ Supply Voltage

= 50% $\times$ 5V

= 2.5V

This does **not** mean that the pin is actually producing 2.5V continuously. Instead, the load behaves **as if** it were receiving approximately 2.5V.

Different loads respond differently to this rapidly changing signal.

Why Different Loads Behave Differently

An interesting question naturally arises. If the output is repeatedly turning ON and OFF, why don't we notice it? The answer depends upon the device connected to the PWM output.

LEDs

Human vision cannot detect extremely fast flashing. If an LED flashes thousands of times every second, our eyes perceive only its average brightness. This principle is also used in modern television displays, computer monitors, and smartphone screens.

DC Motors

Electric motors possess mechanical inertia. Even if power is interrupted for a tiny fraction of a millisecond, the motor continues rotating. The motor therefore averages the applied energy and rotates at a speed roughly proportional to the PWM duty cycle.

Heating Elements

A heater does not cool instantly when power is removed. Its thermal mass smooths the rapid ON/OFF switching. The average heat generated depends on the average power delivered.

Audio Speakers

A speaker cone has mass and elasticity. Rapid PWM pulses can be filtered electrically or mechanically to produce audible sound. Many inexpensive audio amplifiers use PWM internally because it provides excellent efficiency.

PWM is Everywhere

Most people use PWM every day without realizing it.

Examples include:

- LED room lighting
- Laptop display brightness
- Mobile phone screens
- Computer cooling fans
- Electric vehicles
- Drone motor controllers
- Battery chargers
- Solar charge controllers
- Audio amplifiers
- Microwave oven power control
- Washing machines
- Air conditioners
- Medical equipment
- Industrial automation

In fact, PWM has become one of the most important techniques in modern embedded systems. Whenever energy must be controlled efficiently, there is a good chance that PWM is involved.

Why is it Called "Pulse Width" Modulation?

The name becomes obvious when we look at the waveform. Suppose every cycle lasts exactly the same amount of time.

Only the width of the HIGH pulse changes.

20%



50%

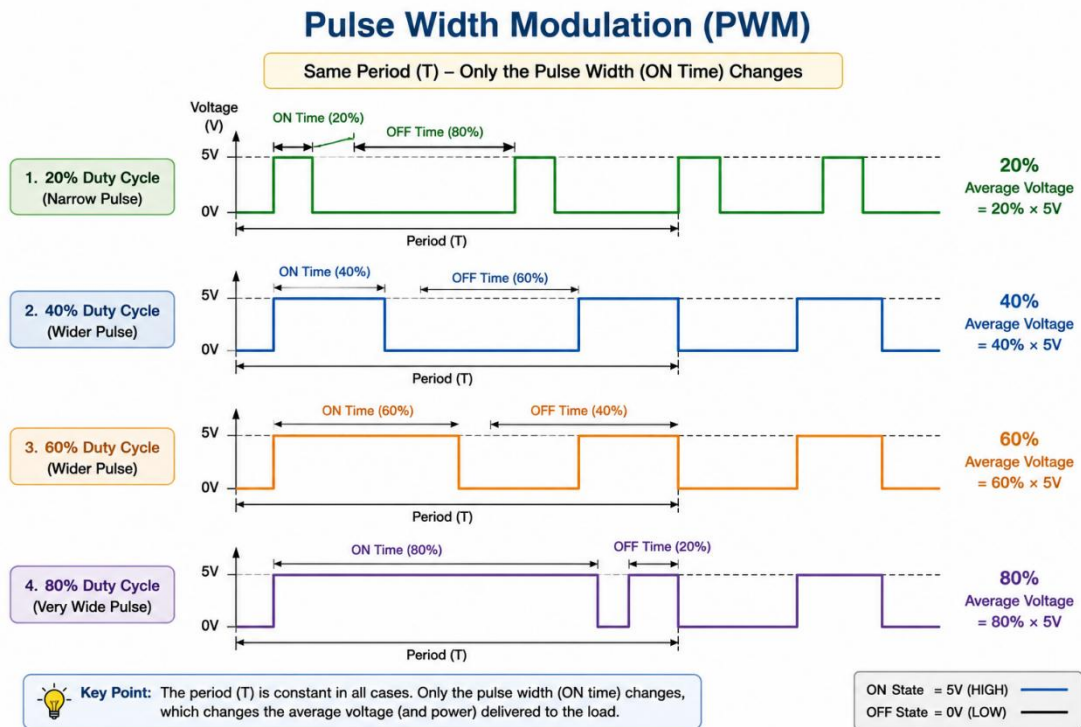


80%



The pulse becomes wider or narrower while the overall period remains unchanged.

Since we are **modulating the pulse width**, the technique is called **Pulse Width Modulation**.



A Powerful Feature Built into AVR Microcontrollers

Generating PWM by software would consume a large amount of CPU time. Fortunately, AVR microcontrollers contain dedicated **hardware timers** capable of generating highly accurate PWM signals automatically.

Once configured, the timer hardware:

- Counts clock pulses
- Generates PWM waveforms
- Maintains precise timing
- Continues operating even while the CPU executes other instructions

This is one of the greatest advantages of hardware PWM.

The processor is free to perform other tasks while the timer continuously generates accurate output waveforms.

In later sections of this chapter, we will learn exactly how these timers work and how the MikroDuino SDK makes them remarkably easy to use.

Why PWM Matters

More Than Just LED Dimming

When beginners first encounter PWM, they usually see a simple demonstration where an LED gradually becomes brighter and dimmer. Although this is an excellent introduction, it barely scratches the surface of what PWM can accomplish. In professional embedded systems, PWM is far more than a method for controlling LED brightness.

It is one of the fundamental techniques used to control **energy, power, motion, light, heat**, and even **sound**. Without PWM, many modern electronic products would either consume far more power or require expensive analog circuitry.

The Challenge of Analog Control

Suppose you want to control the speed of a 12V DC motor. One possible solution is to reduce the voltage using a variable resistor.

Although simple, this approach wastes a significant amount of electrical energy as heat.

Power Supply

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Variable Resistor

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Motor

As the resistor drops voltage, it becomes hot, and also limits the current.

Large motors would require physically large resistors capable of dissipating substantial power. This approach is inefficient and impractical.

PWM Solves This Problem

Instead of reducing voltage, PWM supplies the **full voltage** to the motor—but only for a controlled portion of each cycle.

Battery

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PWM Switch

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Motor

The switching device (usually a MOSFET or transistor) spends very little time in its high-power transition region, making the system highly efficient.

The motor receives pulses of full voltage. Its inertia smooths these pulses into steady rotation. The result is excellent speed control with minimal wasted energy.

Why Engineers Love PWM

PWM offers numerous practical advantages.

High Efficiency

Because the switching transistor is usually either fully ON or fully OFF, very little energy is wasted. Efficiencies exceeding **90–95%** are common in PWM-based systems.

Low Heat Generation

Reduced power loss means less heat. Smaller heat sinks can be used, making products lighter, cheaper, and more reliable.

Precise Digital Control

Since PWM is generated digitally, software can adjust the duty cycle with exceptional precision. Brightness, speed, temperature, and power can all be controlled accurately.

Excellent Repeatability

Unlike analog components that drift with temperature and age, digital PWM signals remain highly repeatable. The same duty cycle produces consistent results every time.

Easy Integration with Sensors

PWM works naturally with sensor feedback.

For example:

Temperature Sensor

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Microcontroller

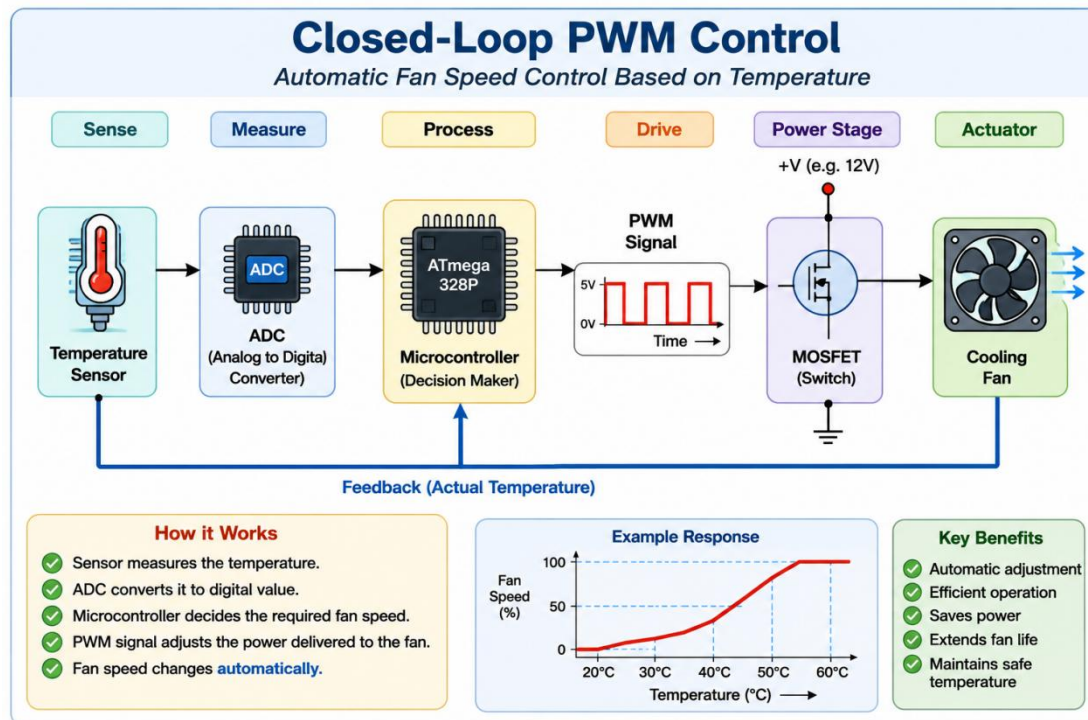
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PWM Output

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Cooling Fan

As temperature rises, software simply increases the PWM duty cycle. No complex analog circuitry is required.



Common Embedded Applications of PWM

PWM appears in nearly every field of embedded electronics.

Lighting Control

- LED brightness adjustment
- RGB color mixing
- Stage lighting
- Automotive lighting
- Decorative lighting

Motor Control

- DC motors
- Brushless motors (BLDC)
- Cooling fans
- Robotics
- Conveyor systems
- Electric bicycles
- Electric vehicles

Heating Control

- 3D printers
- Soldering stations
- Water heaters
- Laboratory incubators
- Temperature-controlled ovens

Power Electronics

PWM forms the heart of modern power converters:

- Buck converters
- Boost converters
- Battery chargers
- Solar charge controllers
- Uninterruptible power supplies (UPS)
- Inverters

Audio Systems

PWM can also generate sound (Class D Amplifiers)

Applications include:

- Buzzers
- Electronic musical instruments
- Digital amplifiers
- Speech generation
- Tone generation

Waveform Generation

By continuously varying the PWM duty cycle according to mathematical values stored in memory, it is possible to generate:

- Sine waves
- Triangle waves
- Sawtooth waves
- Arbitrary custom waveforms

After passing through a simple low-pass filter, these PWM signals can closely approximate analog waveforms.

Later in this chapter, we will build a PWM-based waveform generator using the MikroDuino SDK.

PWM and Embedded Systems

Among all the peripherals inside an AVR microcontroller, PWM is arguably one of the most versatile.

A single timer can simultaneously:

- Control LED brightness
- Regulate motor speed
- Position a servo motor
- Generate musical tones
- Produce precise timing signals
- Drive switching power supplies

This extraordinary flexibility explains why nearly every modern microcontroller—from small 8-bit AVR devices to powerful 32-bit ARM processors—includes dedicated hardware PWM modules.

For an embedded systems engineer, understanding PWM is not simply another programming skill—it is a foundational concept that unlocks efficient, precise, and intelligent control of the physical world.